THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE FULL SYLLABUS PAPER SET 01

Full Syllabus (Chapters 1–11) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The acid present in vinegar is: 1

(a) formic acid **(b)** citric acid **(c)** acetic acid **(d)** tartaric acid

2. A solution turns red litmus blue. It is: 1

(a) sour **(b)** basic **(c)** acidic **(d)** neutral

3. Which of these should never be touched with wet hands? 1

(a) Switch connected to mains **(b)** A notebook

(c) A book **(d)** A pencil

4. The material of a bulb's filament is: 1

(a) tungsten **(b)** iron **(c)** copper **(d)** aluminium

5. The process of formation of a layer of oxide on metals is called: 1

(a) corrosion **(b)** distillation **(c)** sublimation **(d)** filtration

6. The metal that does not rust easily and is used on bridges and utensils: 1

(a) plain iron **(b)** sodium **(c)** stainless steel **(d)** copper sulfate

7. Which of these is a chemical change? 1

(a) Boiling water **(b)** Cutting paper **(c)** Cooking an egg **(d)** Melting wax

8. Lime water turns milky with: 1

(a) hydrogen **(b)** oxygen **(c)** carbon dioxide **(d)** nitrogen

9. Spinach, kidney beans, raisins and figs are rich in: 1

(a) vitamin C **(b)** iron **(c)** iodine **(d)** fats only

10. The Suvidha Sanitary Napkin initiative provides: 1

(a) free books **(b)** biodegradable pads at affordable prices

(c) free bicycles **(d)** free uniforms

11. Heat transfer in solids mainly takes place by: 1

(a) convection **(b)** radiation **(c)** conduction **(d)** dissolving

12. The water table is: 1

(a) the upper level of groundwater **(b)** the sea level

(c) cloud base **(d)** a table in the lab

13. The time taken by a pendulum to complete one oscillation is its: 1

(a) speed **(b)** frequency **(c)** time period **(d)** amplitude

14. A bus covers 240 km in 4 hours. Its average speed is: 1

(a) 30 km/h **(b)** 960 km/h **(c)** 70 km/h **(d)** 60 km/h

15. In the breathing model, balloons represent: 1

(a) nostrils **(b)** diaphragm **(c)** lungs **(d)** ribs

16. During exhalation the ribs move: 1

(a) down and inwards **(b)** up and outwards **(c)** do not move **(d)** sideways only

17. Mushrooms obtain food from: 1

(a) soil water **(b)** dead and decaying matter

(c) air **(d)** sunlight

18. Pot C kept in the dark with water shows poor growth because of lack of: 1

(a) soil **(b)** air **(c)** water **(d)** sunlight

19. A transparent object forms: 1

(a) a dark shadow **(b)** luminous glow **(c)** almost no shadow **(d)** a clear image

20. Different constellations are seen in different seasons because of Earth's: 1

(a) revolution **(b)** tilt only **(c)** rotation **(d)** eclipse

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 15–25 words each.

21. What happens to red rose extract when lemon juice is added? 2

22. Why is it dangerous to touch electric devices with wet hands? 2

23. What is an alloy? Give an example. 2

24. Name two foods rich in calcium and two rich in iron. 2

25. Name two good and two poor conductors of heat. 2

26. What does an odometer measure? 2

27. What are alveoli? What is their function? 2

28. Why are leaves usually flat and thin? 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. Name five components of a torch and give their functions. 3

30. Distinguish between rusting and burning. 3

31. How does the Sun drive the water cycle? 3

32. Why does a fish die out of water although air has more oxygen? 3

33. List the activities to classify materials as transparent, translucent or opaque. 3

34. Why is the pole star useful? 3

35. A solution turns red cabbage juice green. What is its nature and one example? 3

OR

A field has become acidic. Name the substance a farmer should add and give the reason. 3

36. Why must the leaf be boiled in alcohol before testing? 3

OR

Explain why plants are called autotrophs. 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 80–120 words each.

37. **Case study:** A ship's hull is rusting due to salty water. 5

(i) Why does sea water speed it up?
(ii) What coating?
(iii) Which metal can be fixed to it?
(iv) What is that protection called?

OR

Case study: A student places a nail, a copper wire, a plastic straw and a pencil lead in a circuit tester. 5

(i) Which light the bulb?
(ii) Which do not?
(iii) Why does lead conduct?
(iv) State a use of the conductors.

38. (i) Explain secondary sexual characteristics in boys and girls. 5

(ii) Why are they called so?
(iii) What are the primary ones?

OR

(i) List five points for making adolescence a joyful experience. 5

(ii) Explain two.

39. (i) Write the relation among speed, distance and time. 5

(ii) A train takes 4 h to go 320 km. Find speed.
(iii) How long to cover 480 km?

OR

Case study: A distance–time table: 0 s 0 m; 10 s 50 m; 20 s 100 m; 30 s 160 m. 5

(i) Is it uniform?
(ii) Speed for first 20 s.
(iii) Speed for last 10 s.
(iv) Overall average speed.

40. (i) What is lateral inversion? 5

(ii) Why on an ambulance?
(iii) Another example of its use.

OR

(i) How does the shadow of a person change from morning to noon? 5

(ii) Why?
(iii) Use in sundials.

END OF QUESTION PAPER Total: 80 Marks

MASTERMIND